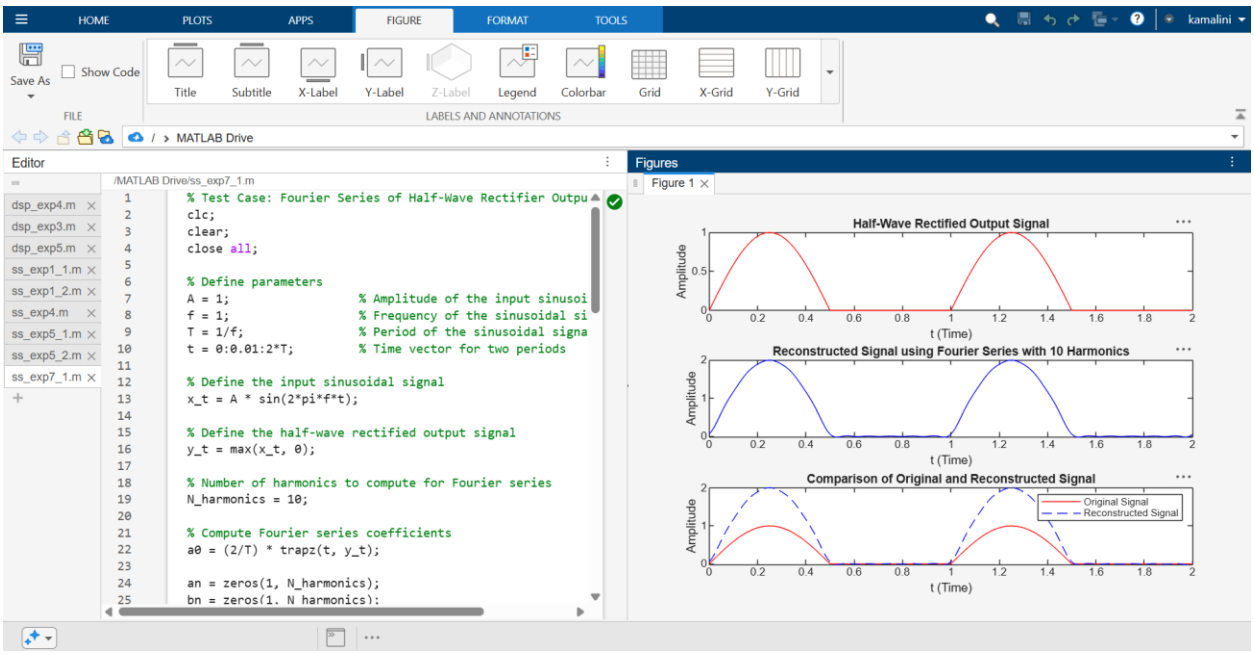
EXPERIMENT 7

```
% Test Case: Fourier Series of Half-Wave Rectifier Output
clc;
clear;
close all;
% Define parameters
A = 1; % Amplitude of the input sinusoidal signal
f = 1; % Frequency of the sinusoidal signal (Hz)
T = 1/f; % Period of the sinusoidal signal
t = 0:0.01:2*T; % Time vector for two periods
% Define the input sinusoidal signal
x_t = A * sin(2*pi*f*t);
% Define the half-wave rectified output signal
y_t = max(x_t, 0);
% Number of harmonics to compute for Fourier series
N_harmonics = 10;
% Compute Fourier series coefficients
a0 = (2/T) * trapz(t, y_t);
an = zeros(1, N_harmonics);
bn = zeros(1, N_harmonics);
for n = 1:N_harmonics
an(n) = (2/T) * trapz(t, y_t .* cos(2*pi*n*f*t));
bn(n) = (2/T) * trapz(t, y_t .* sin(2*pi*n*f*t));
end
% Reconstruct the signal using Fourier series
y_reconstructed = (a0/2) * ones(size(t));
for n = 1:N_harmonics
y_reconstructed = y_reconstructed ...
+ an(n) * cos(2*pi*n*f*t) ...
+ bn(n) * sin(2*pi*n*f*t);
end
% Plot the original half-wave rectified signal
figure;
clf;
subplot(3,1,1);
plot(t, y_t, 'r');
xlabel('t (Time)');
ylabel('Amplitude');
title('Half-Wave Rectified Output Signal');
% Plot the reconstructed signal using Fourier series
subplot(3,1,2);
plot(t, y_reconstructed, 'b');
xlabel('t (Time)');
ylabel('Amplitude');
title(['Reconstructed Signal using Fourier Series with ', ...
num2str(N_harmonics), ' Harmonics']);
```

```
% Plot the original vs reconstructed signal for comparison
subplot(3,1,3);
plot(t, y_t, 'r', t, y_reconstructed, 'b--');
xlabel('t (Time)');
ylabel('Amplitude');
title('Comparison of Original and Reconstructed Signal');
legend('Original Signal', 'Reconstructed Signal');
```



2:

% Test Case: Fourier Series of Half-Wave Rectifier Output

clc; clear; close all;

% Define parameters A = 1;

% Amplitude of the input sinusoidal signal f = 1;

% Frequency of the sinusoidal signal (Hz) T = 1/f;

% Period of the sinusoidal signal t = 0:0.01:2*T;

% Time vector for two periods

% Define the input sinusoidal signal x_t = A * sin(2*pi*f*t);

% Define the half-wave rectified output signal y_t = max(x_t, 0);

```
% Number of harmonics to compute for Fourier series N_harmonics = 10;

% Compute Fourier series coefficients a0 = (2/T) * trapz(t, y_t);

an = zeros(1, N_harmonics); bn = zeros(1, N_harmonics);

for n = 1:N_harmonics an(n) = (2/T) * trapz(t, y_t .* cos(2*pi*n*f*t)); bn(n) = (2/T) * trapz(t, y_t .* sin(2*pi*n*f*t)); end

% Reconstruct the signal using Fourier series y_reconstructed = (a0/2) * ones(size(t));

for n = 1:N_harmonics y_reconstructed = y_reconstructed ... + an(n) * cos(2*pi*n*f*t) ... + bn(n) * sin(2*pi*n*f*t); end

% Plot the original half-wave rectified signal figure;

 clf;

subplot(3,1,1); plot(t, y_t, 'r');

 xlabel('t (Time)');

 ylabel('Amplitude');

 title('Half-Wave Rectified Output Signal');

% Plot the reconstructed signal using Fourier series subplot(3,1,2);

plot(t, y_reconstructed, 'b');

xlabel('t (Time)');

 ylabel('Amplitude');

 title(['Reconstructed Signal using Fourier Series with ', ... num2str(N_harmonics), ' Harmonics']);

% Plot the original vs reconstructed signal for comparison subplot(3,1,3);

plot(t, y_t, 'r', t, y_reconstructed, 'b--');

 xlabel('t (Time)');

ylabel('Amplitude');
```

title('Comparison of Original and Reconstructed Signal');

legend('Original Signal', 'Reconstructed Signal');

